

Selection of Data-Plane Rules for Controller Escalation

Objective

The objective is to retain simple flow checks in the programmable data plane and transfer only a small, informative subset of flows to the controller for detailed analysis. The target considered here is a controller-bound rate of at most 1% of all flows. Each selected flow should generate only one digest or notification; transferring every packet of a selected flow would not satisfy the same traffic bound.

Feature definitions

Symbol	Feature	Meaning
PLM	Packet Length Max	Largest packet length observed in the flow
PSH	PSH Flag Count	Number of packets carrying the TCP PSH flag
TLF	Total Length of Fwd Packet	Cumulative bytes in the forward direction
FIN	FIN Flag Count	Number of packets carrying the TCP FIN flag
BPKT	Total Bwd packets	Number of packets observed in the backward direction

How the rules are derived

A trained decision tree classifies a flow by following comparisons from its root to a leaf. To derive a rule, we select a leaf labeled as an attack and collect every comparison along its root-to-leaf path. Taking a left branch contributes $x_f \leq t$, whereas taking a right branch contributes $x_f > t$. Conditions involving the same feature are combined into a single interval. Because these features are integer-valued, fractional thresholds are converted into equivalent integer bounds. For example, $PSH > 4.5$ and $PSH \leq 11.5$ becomes $5 \leq PSH \leq 11$.

$$R(x) = \bigwedge_{f \in P} \{ x_f \leq t \text{ for a left branch; } x_f > t \text{ for a right branch} \}$$

Rule	PLM	PSH	TLF	FIN	BPKT	Tree prediction
1	≤ 34	5–11	28–128	Any	Any	FTP-Patator
2	≥ 4342	≤ 13	≤ 128	Any	Any	PortScan
3	≤ 964	2–10	195–541	≤ 1	≤ 3	DoS-Slowloris
4	231–232	2–10	≥ 129	≤ 1	$= 4$	DoS-Slowloris
5	255–961	≤ 13	≥ 2823	≤ 1	6–8	DoS-Slowloris
6	≤ 964	≤ 13	≥ 542	≥ 2	≤ 3	DoS-Slowloris

Six selected attack-path rules: The combined decision is $ESCALATE(x) = R_1(x) \vee R_2(x) \vee \dots \vee R_6(x)$. A flow matching multiple rules is counted and reported only once. The model exports 71 attack-leaf paths. These six are selected representative paths: one FTP-Patator path, one PortScan path, and four DoS-Slowloris paths. They are not the complete classifier and were not originally produced by an optimization algorithm.

Selecting rules under a 1% controller budget

Let $\mathcal{R}$ denote all candidate rules, $\mathcal{M}_r$ the set of flows matched by rule r , $\mathcal{A}$ the set of labeled attack flows, and N the total number of flows. The desired subset maximizes the number of covered attacks while limiting the union of matched flows to 1%.

$$S^* = \arg \max_{\mathcal{S} \subseteq \mathcal{R}} |\mathcal{A} \cap (\bigcup_{r \in \mathcal{S}} \mathcal{M}_r)|$$

subject to $|\bigcup_{r \in \mathcal{S}} \mathcal{M}_r| / N \leq 0.01$

The union ensures that a flow matching several rules is counted only once. Rule selection should use training data and be evaluated on a separate test split. A data-plane meter is still needed to enforce the hard 1% limit when live traffic differs from the training distribution.

Measured behavior of the six rules

Data split	Total flows	Transferred flows	Transfer rate	Traffic reduction
Training	108,120	108	0.0999%	99.9001%
Held-out test	27,030	26	0.0962%	99.9038%

On the held-out split, the transferred set contains 16 DoS-Slowloris flows, seven PortScan flows, and three FTP-Patator flows. All 26 transferred flows are attacks, but they represent only about 0.155% of the attack flows in that split. The rules are highly selective and remain far below the 1% budget, but they do not provide broad attack coverage (this is based on the >10% of total traffic.)

Using the remaining budget and monitoring zero-day behavior

A zero-day flow can fall within a region associated with common benign traffic because these rules rely on only a small set of aggregated features. For the held-out split, a strict 1% budget permits 270 transfers. The six rules consume 26 transfers, leaving room for 244 additional flows. Sampling 244 of the remaining 27,004 flows corresponds to approximately 0.904%.

A practical policy gives priority to the six-rule matches, uses novelty detection or random sampling for non-matching flows, and then applies a global meter:

$$ESCALATE = \text{known-rule match} \vee \text{uncommon signature} \vee \text{sampling common flow},$$

subject to the 1% of total traffic (which can vary according to administration).

This policy preserves the controller budget while providing limited visibility into previously unseen behavior. Sampling improves the probability of observing zero-day activity, although it cannot guarantee detection.

Next steps and escalation policy

Step 1 — Select and validate the rules. Evaluate all 71 attack-leaf rules on the training data, select the subset that maximizes attack-flow coverage under the 1% transfer constraint, and confirm its transfer rate and coverage on held-out data. Because traffic changes over time, repeat this calibration periodically using recent controller observations.

Step 2 — Deploy the escalation policy. For every active flow, the data plane updates the five feature states and evaluates the selected rules. A first-time rule match receives the highest escalation priority. A non-matching flow may still be escalated when its feature signature is uncommon or when it is selected by the background sampler. A per-flow reported bit suppresses duplicate notifications, while a global meter limits the combined controller-bound rate to 1%.

Priority 1: selected attack-rule matches

Priority 2: uncommon or previously unseen feature signatures

The data plane represents each flow using a compact signature:

where each value identifies the range containing the corresponding feature. Frequently observed signatures are stored in a table (K). If a flow's signature is present in (K), the data plane treats it as common and forwards it normally. If the signature is absent, the flow is considered uncommon, and, if the meter permits, the data plane sends a single digest to the controller. An uncommon signature does not necessarily indicate an attack; it only identifies a flow that requires further analysis. The controller updates (K) as traffic changes. It can add signatures that are repeatedly confirmed as safe, remove signatures associated with suspicious behavior, and adjust the table size to control the number of escalated flows. This allows the policy to adapt to recent traffic rather than relying permanently on the original training data.

Priority 3: sampled flows from common regions

A zero-day flow may have the same binned signature as common benign traffic and therefore pass the Priority 2 lookup. Priority 3 provides visibility into this region by sampling flows according to a deterministic hash of the flow identifier. The sampling decision is $\text{SAMPLE}(x) = 1[\text{H}(\text{flowID}(x)) \bmod 10,000 < \tau]$. For example, $\tau = 20$ selects approximately 0.2% of common flows. The decision is made once at flow creation, avoiding the bias toward long flows that would result from sampling individual packets.

Illustrative allocation of the 1% budget

Priority	Purpose	Illustrative budget
1	Selected attack-rule matches	0.10%
2	Uncommon feature signatures	0.70%
3	Sampling within common regions	0.20%
Total	All controller-bound flows	1.00%

The allocation is a configuration choice and must be calibrated on held-out and recent traffic. The six rules consumed approximately 0.096% on the current test split, but this value is not guaranteed for future traffic. A global meter provides the hard aggregate limit, while per-priority meters can reserve capacity for uncommon signatures and benign-region sampling.

Data-plane processing order

1. If a selected attack rule matches, request a Priority 1 escalation.
2. Otherwise, if the feature signature is absent from K, request a Priority 2 escalation.
3. Otherwise, if the flow hash satisfies the sampling condition, request a Priority 3 escalation.
4. Admit the request only when the global meter permits it and the flow has not previously been reported. Otherwise, forward the flow normally without another notification.

For every accepted escalation, the data plane sends a compact digest containing the flow identifier, escalation priority, and reason, matched rule when applicable, current feature bins, and timestamp. The controller performs a deeper analysis and may install a temporary block, monitoring, or forwarding action.

Operational decision

$$\text{ESCALATE}(x) = [\text{RULE_MATCH}(x) \vee \text{UNCOMMON}(x) \vee \text{SAMPLE}(x)]$$

This ordering guarantees that known suspicious patterns are considered first, preserves some visibility into zero-day behavior, prevents repeated reports for the same flow, and keeps the total controller load within the configured budget.

1. Controller load versus attack coverage curve.
- 2.